

Hermes: An Evidence-Driven Agentic Framework for Trustworthy and Explainable AI-Generated Video Detection

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Abstract

Recent advances in generative video models have blurred the boundary between real and synthetic content, raising urgent concerns about digital authenticity. Multimodal large language models (MLLMs) are promising for AI-generated video (AIGV) detection due to their broad perceptual and reasoning capabilities; however, existing MLLM-based detectors remain prone to hallucinated evidence and unstable reasoning, leading to false alarms and generic, unverifiable explanations. To address these issues, we propose **Hermes**, an evidence-driven agentic framework for trustworthy and explainable AIGV detection. Hermes comprises three key components: (1) Adaptive Instance-Conditioned Detection Strategy Planning, (2) Evidence-Centric Reasoning and Verification, and (3) Graph-Grounded Evidence Deliberation. Specifically, Hermes analyzes each video and uses instance-conditioned retrieval-augmented generation to retrieve relevant forensic knowledge and compose a tailored detection strategy. It then constructs a verifiable Evidence Reasoning Graph (ERG) to keep the reasoning grounded in concrete video evidence. Finally, multi-agent deliberation audits and refines the ERG to reconcile conflicting evidence and improve the reliability of the final judgment. Together with a library of forensic tools, these components enable evidence-grounded authenticity judgments and structured, verifiable explanations. Extensive experiments show that Hermes, without task-specific training, achieves state-of-the-art performance and produces auditable explanations.

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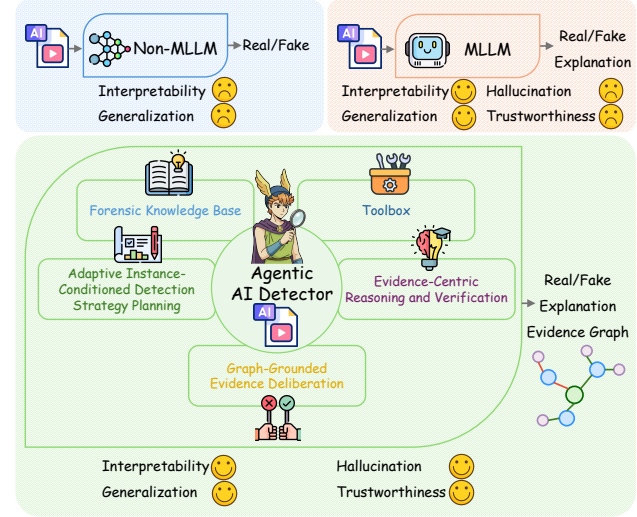


Figure 1. **Comparison with prior paradigms.** Conventional non-MLLM detectors offer limited interpretability and generalization, whereas direct MLLM-based detectors provide explanations but remain vulnerable to hallucination and unreliable reasoning. Hermes reframes detection as evidence-driven agentic reasoning, enabling interpretable and generalizable authenticity assessment while mitigating hallucination and improving trustworthiness.

1. Introduction

The rapid advancement of generative models, particularly diffusion-based frameworks (Dhariwal & Nichol, 2021; Lu et al., 2023), has substantially improved the realism and controllability of video synthesis (Li et al., 2026a; Ho et al., 2022). While these advances have benefited content creation, entertainment, and simulation (Chen et al., 2025; Li et al., 2026b; Zhang et al., 2025b), they have also raised serious concerns about digital media authenticity and potential misuse (Li et al., 2025b; Lu et al., 2025; 2024). Consequently, developing reliable and interpretable detection methods has become a critical challenge in synthetic video forensics (Li et al., 2021).

Recent studies (Song et al., 2024; Zhang et al., 2025c; Li et al., 2024) have explored spatial and temporal cues for AI-generated video detection, showing encouraging performance. However, many existing models provide limited insight into the evidence underlying their predictions, mak-

ing their decisions difficult to interpret or audit (Wen et al., 2025a). Meanwhile, multimodal large language models (MLLMs) have demonstrated strong perceptual and reasoning capabilities (Wu et al., 2023; Schulze Buschoff et al., 2025), motivating their use in multimedia forensics. Early studies have introduced MLLMs into image forgery detection, achieving promising results (Xu et al., 2025; Zhou et al., 2025). Nevertheless, direct MLLM inference remains susceptible to hallucination and unstable reasoning (Tonmoy et al., 2024). These limitations are further amplified in video settings by greater visual complexity, temporal dynamics, and richer semantics (Ma et al., 2024; Li et al., 2026c; Sun et al., 2025), leading to unreliable authenticity judgments and explanations that lack concrete evidential support, as illustrated in Figure 1.

We argue that a central challenge is that MLLMs reason within an unconstrained semantic space rather than a video-grounded, authenticity-relevant forensic space. First, MLLMs may overemphasize high-level semantic content unrelated to authenticity assessment, such as scene narratives, visual aesthetics, or emotional atmosphere, while overlooking concrete forensic evidence (Ma et al., 2024). Second, even when focusing on forgery-related cues, MLLMs may select forensic checks that do not match the video content. For example, they may perform human-anatomy checks when no person is present (Sun et al., 2024). Third, free-form single-pass inference lacks explicit evidence organization and review, allowing unsupported or conflicting findings to directly influence the final judgment (Li et al., 2025a). Together, these limitations raise a question: How can MLLMs make accurate authenticity judgments while providing structured and verifiable explanations?

To address this challenge, we seek to transform open-ended MLLM inference into a controlled forensic process grounded in the input video. We therefore propose **Hermes**, an evidence-driven agentic framework for trustworthy and explainable AIGV detection. Inspired by how human forensic investigators systematically examine cases, select appropriate analyses, and revise conclusions as evidence evolves, Hermes adaptively analyzes each video and identifies the forensic checks relevant to its content. It then performs evidence-centric reasoning with the corresponding tools, organizes the resulting findings into an Evidence Reasoning Graph (ERG), and iteratively audits and refines the graph through multi-agent deliberation. Through this process, Hermes grounds its authenticity judgments in concrete, video-specific forensic evidence and provides structured explanations whose supporting evidence can be traced and verified.

As shown in Figure 2, Hermes comprises three tightly coupled components that progressively constrain the reasoning scope, structure the supporting evidence, and verify the re-

sulting judgment. First, **Adaptive Instance-Conditioned Detection Strategy Planning** analyzes the input video and retrieves relevant forensic knowledge and corresponding tools from a structured knowledge base via retrieval-augmented generation (RAG) (Lewis et al., 2020). It then composes a video-specific detection strategy that tailors the subsequent forensic analysis to the content of the input video. Second, **Evidence-Centric Reasoning and Verification** draws on this strategy to guide tool invocation and evidence analysis, organizing the resulting forensic findings into an explicit ERG. The ERG structures the evidence underlying the judgment, making its provenance and contribution to the authenticity judgment explicit and verifiable. Finally, **Graph-Grounded Evidence Deliberation** introduces an iterative multi-agent process that audits and refines the ERG. The Primary Deliberator supplements missing or uncertain evidence, the Counter Deliberator challenges unreliable findings and relations, and the Consensus Deliberator adjudicates the proposed revisions and determines whether another deliberation round is required. Notably, Hermes requires no task-specific training and can be directly applied across diverse video generators without retraining.

Our main contributions are summarized as follows:

- We present **Hermes**, an evidence-driven agentic framework for AI-generated video detection that adaptively plans forensic analysis and organizes video-specific evidence. This framework enables reliable authenticity judgments and structured, verifiable explanations within a unified forensic reasoning process.
- We develop three coordinated technical components: Adaptive Instance-Conditioned Detection Strategy Planning, which constructs a video-specific detection strategy from video context and retrieved forensic knowledge; Evidence-Centric Reasoning and Verification, which structures forensic evidence within an Evidence Reasoning Graph to make the reasoning process explicit; and Graph-Grounded Evidence Deliberation, which iteratively audits and refines the structured evidence through multi-agent deliberation.
- Extensive experiments across diverse video generators demonstrate that Hermes consistently outperforms state-of-the-art methods in detection performance and cross-generator generalization, while reducing false alarms and missed detections for more trustworthy authenticity judgments.

2. Related Work

2.1. AI-Generated Video Detection

Early synthetic video detectors primarily targeted facial deepfakes, exploiting face-specific priors such as physiological anomalies (Li et al., 2018), spatio-temporal arti-

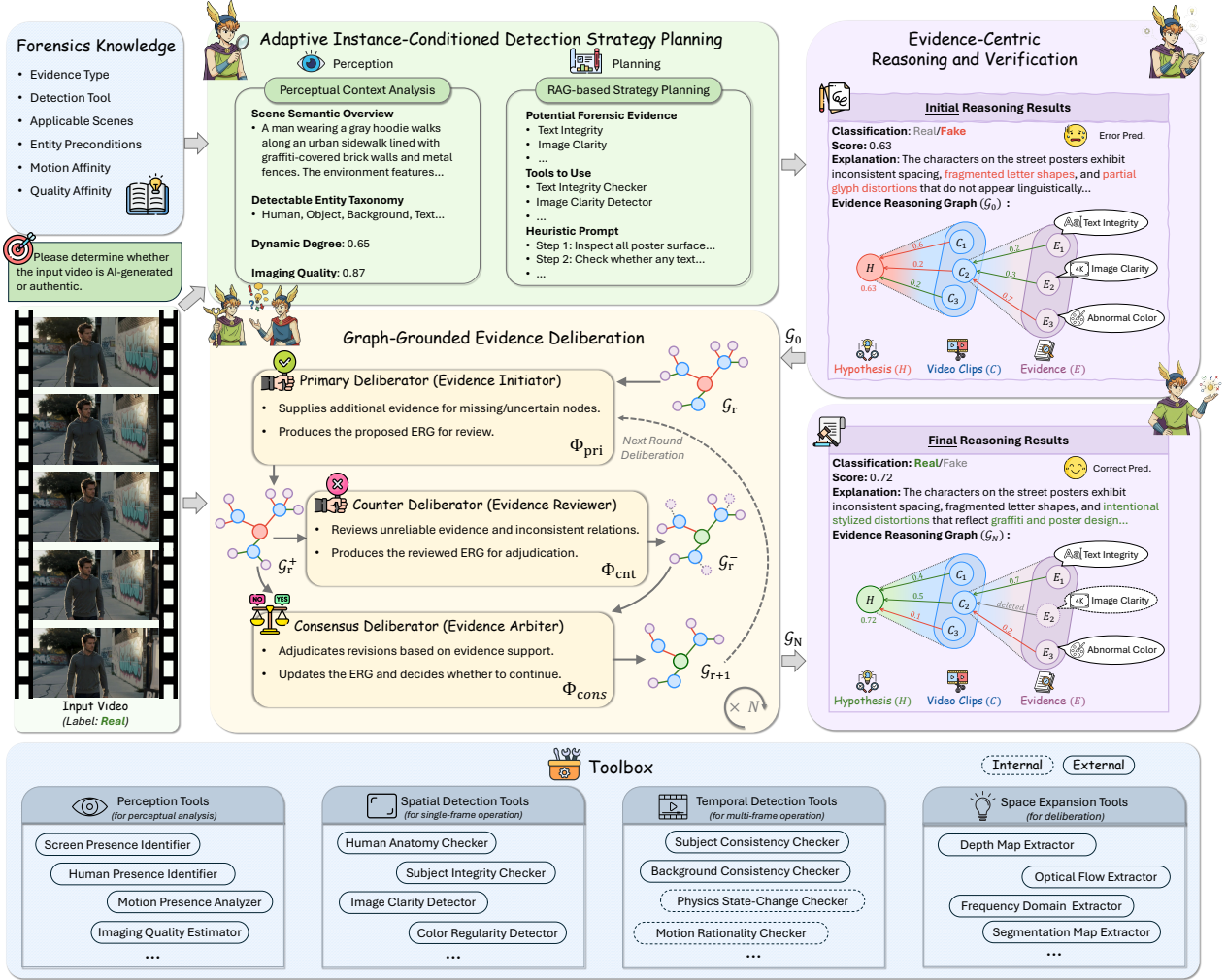


Figure 2. **Overview of the Hermes framework.** Given an input video, Hermes first analyzes its perceptual context and combines it with retrieved forensic knowledge to construct a video-specific detection strategy. Guided by this strategy, Hermes invokes appropriate tools to acquire evidence and organizes the resulting findings into an initial Evidence Reasoning Graph (ERG). Multi-agent deliberation then supplements missing evidence, challenges unreliable findings and relations, and adjudicates proposed revisions, progressively refining the graph into a coherent and verifiable basis for trustworthy authenticity assessment.

facts (Concas et al., 2024), and multi-view facial representations (Peng et al., 2024). Although effective in constrained, face-centric settings, their reliance on facial priors limits their applicability to fully AI-generated videos with diverse subjects and scenes (Kundu et al., 2025). With rapid advances in video generation, recent studies have shifted toward general-purpose AIGV detection. Some methods combine frame appearance with optical flow or motion residuals to capture spatio-temporal inconsistencies (Song et al., 2024; Bai et al., 2024), while others employ Transformer-based architectures (Kundu et al., 2025) or Mamba-style architectures (Chen et al., 2024) to model long-range dependencies. Additionally, some methods explore explicit physical priors and statistical cues, including NSG-VD (Zhang et al., 2025c), which incorporates motion and physical regularities, and D³ (Zheng et al., 2025), which performs training-free de-

tection using second-order statistics. Despite these advances, existing methods typically rely on fixed forensic pipelines across videos and often lack structured, inspectable evidence, limiting their ability to adapt to video-specific content, motion, and imaging quality. In contrast, we pursue video-conditioned forensic planning and structured evidence reasoning to adapt the forensic process to each input video.

2.2. MLLM-Based Forgery Detection

MLLMs have recently been explored for media forensics owing to their broad perceptual, reasoning, and cross-modal understanding capabilities (Ye et al., 2025). Earlier media-forensics studies examined synthetic speech and machine-generated text, where modality-specific artifacts provide useful forensic cues (Přibil et al., 2017; Liu et al., 2022a).

With advances in vision-language models, recent studies have increasingly explored visual forgery detection (Gao et al., 2025; Huang et al., 2025), covering tasks such as authenticity assessment, suspicious-region localization, and explanation generation. FakeShield (Xu et al., 2025), for example, jointly predicts authenticity and produces region-aware explanations, while BusterX++ (Wen et al., 2025b) develops a unified MLLM framework for detecting and explaining AI-generated visual content. Despite this progress, existing MLLM-based detectors often rely on single-pass, free-form reasoning, with limited mechanisms to organize, verify, or revise the evidence supporting their predictions. Their explanations may therefore remain generic or weakly grounded, while the reasoning process remains susceptible to hallucination (Guan et al., 2024; Qi et al., 2026a;b). In contrast, we organize video evidence into an explicit reasoning graph and iteratively audit and refine the structured evidence through graph-grounded deliberation, coupling authenticity judgments with inspectable explanations.

3. Method

As illustrated in Figure 2, Hermes is an evidence-driven agentic framework for trustworthy and explainable AI-generated video detection. It reformulates the task as a structured, video-conditioned reasoning process. The framework comprises three tightly coupled components: (1) Adaptive Instance-Conditioned Detection Strategy Planning, (2) Evidence-Centric Reasoning and Verification, and (3) Graph-Grounded Evidence Deliberation. Together, these components adapt forensic analysis to each video, organize the resulting forensic findings into an Evidence Reasoning Graph, and iteratively audit and refine the structured evidence to produce the final authenticity judgment.

3.1. Adaptive Instance-Conditioned Detection Strategy Planning

Different videos exhibit diverse visual contexts and authenticity-related cues, limiting the effectiveness of a fixed detection pipeline (Liu et al., 2022b; Zhang et al., 2023). To address this challenge, Hermes introduces an instance-conditioned planning module that analyzes each input video before evidence extraction, identifies potentially relevant forensic evidence types, and recommends suitable tools. By accounting for the content and characteristics of each video, the module guides subsequent reasoning toward authenticity-relevant forensic checks and reduces drift toward irrelevant cues.

Forensic Knowledge Base and Toolbox. To support targeted forensic analysis, Hermes organizes its toolbox into four functional groups. Perception Tools $\mathcal{T}_{\text{perc}}$ characterize the input video during perceptual context analysis; Spatial Detection Tools $\mathcal{T}_{\text{spa}}$ and Temporal Detection Tools $\mathcal{T}_{\text{temp}}$

examine frame-level and cross-frame cues, respectively, to extract forensic evidence; and Space Expansion Tools $\mathcal{T}_{\text{exp}}$ derive complementary representations, such as depth maps, for further examination during deliberation. Across the four functional groups, tools are also categorized by implementation as internal or external: internal tools rely on structured question-answering procedures, whereas external tools invoke off-the-shelf vision models.

To facilitate video-specific forensic planning, we construct a structured Forensic Knowledge Base that organizes spatial and temporal evidence types together with their corresponding detection tools. Specifically, based on existing forensic taxonomies (Zhang et al., 2025a; Liu et al., 2025), we define 10 temporal evidence types (e.g., Subject Consistency) and 9 spatial evidence types (e.g., Human Anatomy and Text Integrity). We denote these evidence types and their associated detection tools by two index-aligned collections, $\mathcal{F} = \{\epsilon_i\}_{i=1}^M$ and $\mathcal{T}_{\text{det}} = \mathcal{T}_{\text{spa}} \cup \mathcal{T}_{\text{temp}} = \{\tau_i\}_{i=1}^M$, where each τ_i is the spatial or temporal detection tool associated with evidence type ϵ_i .

To support context-aware retrieval, we use an MLLM offline to organize the predefined evidence-tool pairs into structured knowledge entries:

$$\mathcal{K} = \Phi_{\text{MLLM}}(\mathcal{F}, \mathcal{T}_{\text{det}}, \mathcal{P}_{\text{know}}), \quad (1)$$

where $\mathcal{P}_{\text{know}}$ specifies the entry schema and output constraints. Each entry in $\mathcal{K}$ contains six attributes: *Evidence Type*, *Detection Tool*, *Applicable Scenes*, *Entity Preconditions*, *Motion Affinity*, *Quality Affinity*. The last four attributes serve as retrieval fields for matching the perceptual context of an input video with relevant evidence-tool pairs.

Perceptual Context Analysis. To characterize each input video for subsequent forensic planning, Hermes first constructs a structured perceptual context. Given a video V , a vision-language model (VLM), guided by the scene-oriented prompt $\mathcal{P}_{\text{scene}}$, analyzes the video and can selectively invoke applicable tools from $\mathcal{T}_{\text{perc}}$ as needed:

$$\mathcal{C}_{\text{perc}} = \Phi_{\text{VLM}}(V, \mathcal{P}_{\text{scene}}). \quad (2)$$

The resulting context $\mathcal{C}_{\text{perc}}$ comprises four attributes. *Scene semantics* describes the overall scene and observed content, while *entity taxonomy* identifies the observed entity types in the video. *Motion dynamics* characterizes the degree and pattern of temporal variation, and *imaging quality* assesses visual conditions such as clarity, resolution, and noise. Together, these attributes provide a compact, video-specific context for subsequent strategy planning.

RAG-based Strategy Planning. To compose a video-specific detection strategy while constraining open-ended reasoning, Hermes performs retrieval-augmented planning

over the perceptual context and structured forensic knowledge. Given $\mathcal{C}_{\text{perc}}$ and the Forensic Knowledge Base $\mathcal{K}$, Hermes first retrieves the top- k evidence-tool entries most relevant to the input video:

$$\mathcal{K}^* = \text{RetrieveTopK}(\mathcal{C}_{\text{perc}}, \mathcal{K}). \quad (3)$$

The MLLM planner then integrates the retrieved subset $\mathcal{K}^*$ with the perceptual context $\mathcal{C}_{\text{perc}}$ to generate a video-specific detection strategy:

$$\begin{aligned} \mathcal{S}_{\text{plan}} &= \Phi_{\text{MLLM}}(\mathcal{K}^*, \mathcal{C}_{\text{perc}}, \mathcal{P}_{\text{plan}}) \\ &= \{\mathcal{F}^*, \mathcal{T}_{\text{det}}^*, \Pi_{\text{heu}}\}. \end{aligned} \quad (4)$$

Here, $\mathcal{P}_{\text{plan}}$ specifies the strategy-generation requirements. $\mathcal{F}^*$ contains the forensic evidence types selected for the input video, while $\mathcal{T}_{\text{det}}^*$ contains the corresponding tools recommended for subsequent analysis. The heuristic prompt Π_{heu} provides structured guidance for examining these evidence types. Together, these elements focus subsequent reasoning on video-relevant forensic checks, supporting targeted and interpretable evidence analysis.

3.2. Evidence-Centric Reasoning and Verification

To ground video authenticity judgments in explicit and inspectable forensic findings, Hermes performs Evidence-Centric Reasoning and Verification. Drawing on the adaptive strategy $\mathcal{S}_{\text{plan}}$, Hermes conducts video-conditioned forensic analysis across the clips and organizes the resulting findings into an Evidence Reasoning Graph (ERG) $\mathcal{G}$. The ERG links evidence items to their source clips and the global authenticity hypothesis, making the evidential basis of the decision explicit and providing a structured foundation for graph-grounded deliberation.

Initial Evidence-Centric Reasoning. Given an input video V , Hermes partitions it into K temporally ordered clips, $V = \{C_k\}_{k=1}^K$, and treats each clip C_k as the basic unit for evidence extraction and tool invocation. Guided by the adaptive strategy $\mathcal{S}_{\text{plan}}$ and the reasoning prompt $\mathcal{P}_{\text{reason}}$, the VLM is instantiated as the Initial Reasoner Φ_{ini} , which examines each clip, selectively invokes the tools specified in $\mathcal{S}_{\text{plan}}$, and integrates their outputs to produce a structured reasoning result:

$$\begin{aligned} \mathcal{R}_0 &= \Phi_{\text{ini}}(\{C_k\}_{k=1}^K, \mathcal{S}_{\text{plan}}, \mathcal{P}_{\text{reason}}) \\ &= \{r_{\text{cls}}, r_{\text{score}}, r_{\text{expl}}, \mathcal{G}_0\}, \end{aligned} \quad (5)$$

where r_{cls} denotes the predicted authenticity label, r_{score} is its associated confidence, r_{expl} is the evidence-grounded textual explanation, and $\mathcal{G}_0$ is the ERG constructed at this stage. This structured output exposes the specific forensic findings underlying the decision, allowing them to be individually inspected and revised.

Evidence Reasoning Graph (ERG). To explicitly model how forensic findings support or contradict the authenticity

hypothesis, the ERG organizes the resulting evidence into a three-level directed graph comprising evidence nodes, clip nodes, and a global hypothesis node. Specifically, the i -th forensic finding associated with clip C_k is represented as an evidence node:

$$E_{k,i} = (d_{k,i}, s_{k,i}, p_{k,i}, t_{k,i}), \quad (6)$$

where $d_{k,i}$ denotes the evidence descriptor, $s_{k,i}$ denotes its confidence score, $p_{k,i} \in \{\text{Real}, \text{Fake}\}$ denotes its polarity, and $t_{k,i}$ records its temporal span. The evidence nodes associated with clip C_k form the set $\mathcal{E}_k = \{E_{k,i}\}_{i=1}^{N_k}$, where N_k is the number of evidence nodes associated with C_k . The ERG is formulated as:

$$\mathcal{G} = (\{\mathcal{E}_k\}_{k=1}^K, \{C_k\}_{k=1}^K, H, \Gamma), \quad (7)$$

where H denotes the global authenticity hypothesis and Γ encodes the directed relations:

$$\Gamma = \{E_{k,i} \rightarrow C_k, C_k \rightarrow H\}. \quad (8)$$

This graph explicitly links forensic evidence to its source clips and captures how clip-level findings collectively support or contradict the global authenticity hypothesis. By organizing these relations in a structured form, the ERG provides an inspectable basis for subsequent graph-grounded deliberation.

3.3. Graph-Grounded Evidence Deliberation

Although the initial reasoning result is grounded in explicit forensic evidence, its authenticity judgment may still be affected by incomplete, weakly supported, or conflicting findings. To improve its reliability, Hermes introduces a graph-grounded multi-agent deliberation mechanism comprising three complementary roles implemented by the VLM under distinct prompts: the Primary Deliberator (evidence initiator, Φ_{pri}), the Counter Deliberator (evidence reviewer, Φ_{cnt}), and the Consensus Deliberator (evidence arbiter, Φ_{cons}). They supplement missing or uncertain evidence, challenge unreliable findings and relations, and adjudicate the proposed revisions, respectively, thereby iteratively updating the reasoning result and its ERG toward a more consistently supported authenticity judgment.

Starting from the initial ERG $\mathcal{G}_0$, Hermes performs an iterative three-role deliberation loop that progressively refines the graph and its associated reasoning result. Throughout deliberation, the three deliberators are guided by $\mathcal{S}_{\text{plan}}$ and invoke the specified tools as needed. At round r , the Primary Deliberator re-examines the current ERG $\mathcal{G}_r$ and supplements or updates missing and uncertain evidence, producing a proposed ERG $\mathcal{G}_r^+$. The Counter Deliberator then reviews $\mathcal{G}_r^+$, identifies unreliable evidence and inconsistent relations, and proposes targeted revisions to obtain the reviewed ERG

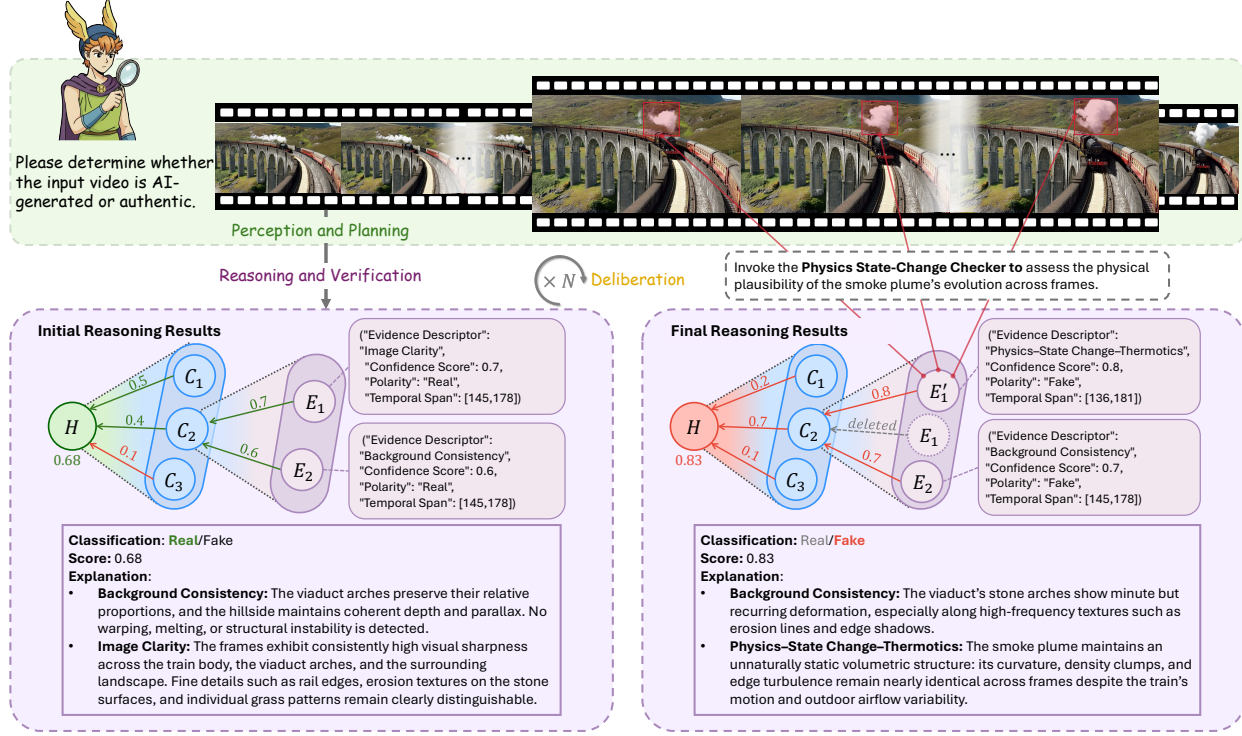


Figure 3. **Graph-grounded deliberation and ERG refinement in Hermes.** Starting from the initial Evidence Reasoning Graph (ERG), Hermes invokes forensic tools during multi-agent deliberation to verify and update the graph. Evidence nodes and relations are supplemented, revised, or removed according to their evidential support, yielding a traceable refinement process from the initial reasoning result to the final detection decision.

$\mathcal{G}_r^-$. The Consensus Deliberator jointly evaluates $\mathcal{G}_r^+$ and $\mathcal{G}_r^-$ based on their evidential support, accepts the valid revisions, and synthesizes the updated ERG $\mathcal{G}_{r+1}$. The updated graph becomes the current ERG for the next round. Deliberation terminates when the Consensus Deliberator accepts no revision and requests no additional evidence, or when the maximum number of deliberation rounds N is reached.

Primary Deliberator. To prevent the reasoning process from being constrained by an incomplete evidential basis, the Primary Deliberator Φ_{pri} re-examines the video with respect to the current ERG $\mathcal{G}_r$ and proposes targeted updates. Guided by the prompt $\mathcal{P}_{\text{pri}}$, it identifies missing or uncertain evidence and supplements or updates the corresponding nodes. Beyond the tools specified in $\mathcal{S}_{\text{plan}}$, it can invoke applicable tools from $\mathcal{T}_{\text{exp}}$ when additional forensic examination is needed. These operations produce a proposed reasoning result:

$$\begin{aligned} \mathcal{R}_r^+ &= \Phi_{\text{pri}}(\{C_k\}_{k=1}^K, \mathcal{G}_r, \mathcal{S}_{\text{plan}}, \mathcal{P}_{\text{pri}}) \\ &= \{r_{\text{cls}}^+, r_{\text{score}}^+, r_{\text{expl}}^+, \mathcal{G}_r^+\}, \end{aligned} \quad (9)$$

where $\mathcal{G}_r^+$ denotes the proposed ERG submitted to the Counter Deliberator for review.

Counter Deliberator. The Counter Deliberator Φ_{cnt} serves as a critical reviewer of the proposed ERG, reassessing its

evidence and relations against the observed video under the critique-oriented prompt $\mathcal{P}_{\text{cnt}}$. Its review operates at two levels: *node reliability* and *structural consistency*. At the node level, it verifies whether individual evidence nodes are sufficiently grounded in the corresponding video observations and revises unreliable attributes when necessary. At the structural level, it adjusts nodes and relations to resolve contradictions and remove unsupported reasoning paths. The resulting revisions are incorporated into a reviewed reasoning result:

$$\begin{aligned} \mathcal{R}_r^- &= \Phi_{\text{cnt}}(\{C_k\}_{k=1}^K, \mathcal{G}_r^+, \mathcal{S}_{\text{plan}}, \mathcal{P}_{\text{cnt}}) \\ &= \{r_{\text{cls}}^-, r_{\text{score}}^-, r_{\text{expl}}^-, \mathcal{G}_r^-\}, \end{aligned} \quad (10)$$

where $\mathcal{G}_r^-$ denotes the reviewed ERG submitted together with $\mathcal{G}_r^+$ to the Consensus Deliberator for adjudication.

Consensus Deliberator. The Consensus Deliberator Φ_{cons} jointly compares the proposed ERG $\mathcal{G}_r^+$ and the reviewed ERG $\mathcal{G}_r^-$ against the observed video. Guided by the adjudication prompt $\mathcal{P}_{\text{cons}}$, it adjudicates the proposed revisions based on evidence grounding and graph consistency:

$$\begin{aligned} \mathcal{R}_r^* &= \Phi_{\text{cons}}(\{C_k\}_{k=1}^K, \mathcal{G}_r^+, \mathcal{G}_r^-, \mathcal{S}_{\text{plan}}, \mathcal{P}_{\text{cons}}) \\ &= \{r_{\text{cls}}^*, r_{\text{score}}^*, r_{\text{expl}}^*, \mathcal{G}_{r+1}\}. \end{aligned} \quad (11)$$

Based on this adjudication, it retains the valid content of

Table 1. Comparison on GenVideo. Recall (R), accuracy (ACC), F1 score (F1), and AUROC of each detector (rows) across the GenVideo subsets (columns), all reported as percentages. Unless otherwise specified, training-based methods follow the common training protocol. For methods marked with †, we use their original training protocols and reported results, while methods marked with ‡ are training-free. The best and second-best results in the Avg. column are marked in **bold** and underlined, respectively.

Methods	Metric	Model Scope	Morph Studio	Moon Valley	HotShot	Show1	Gen2	Crafter	Lavie	Sora	Wild Scrape	Avg.
DeMamba [arXiv 2024]	R	87.00	93.60	98.80	40.60	48.40	98.00	88.40	59.00	48.21	58.20	72.02
	ACC	91.70	95.00	97.60	68.50	72.40	97.20	92.40	77.70	72.32	77.30	84.21
	F1	91.29	94.93	97.63	56.31	63.68	97.22	92.08	72.57	63.53	71.94	80.12
	AUROC	98.04	98.82	99.68	87.84	90.12	99.46	97.81	91.32	88.36	87.38	93.88
NPR [CVPR 2024]	R	61.20	80.00	98.00	16.00	33.00	91.20	80.60	34.60	35.71	43.20	57.35
	ACC	79.80	89.20	98.20	57.20	65.70	94.80	89.50	66.50	67.86	70.80	77.96
	F1	75.18	88.11	98.20	27.21	49.03	94.61	88.47	50.81	52.63	59.67	68.39
	AUROC	93.05	97.18	99.66	82.97	90.50	99.13	97.87	87.54	90.47	91.84	93.02
TALL [ICCV 2023]	R	51.20	65.20	93.40	32.00	61.60	94.80	81.80	49.20	25.00	53.60	60.78
	ACC	75.10	82.10	96.20	65.50	80.30	96.90	90.40	74.10	61.61	76.30	79.85
	F1	67.28	78.46	96.09	48.12	75.77	96.83	89.50	65.51	39.44	69.34	72.63
	AUROC	95.82	97.14	99.73	92.55	97.36	99.79	99.09	94.84	86.67	93.75	95.67
STIL [ACMMM 2021]	R	73.80	70.80	43.40	1.40	2.00	45.00	13.20	7.20	1.79	11.60	27.02
	ACC	86.90	85.40	71.70	50.70	51.00	72.50	56.60	53.60	50.89	55.80	63.51
	F1	84.93	82.90	60.53	2.76	3.92	62.07	23.32	13.43	3.51	20.79	35.82
	AUROC	96.43	97.77	99.34	86.66	90.56	98.88	97.04	88.16	92.57	87.52	93.49
NSG-VD [NeurIPS 2025]	R	68.33	98.33	100.00	92.50	87.50	80.00	98.33	94.17	78.57	82.50	88.02
	ACC	81.67	98.33	96.67	91.67	90.83	88.33	95.83	94.17	88.39	88.75	91.46
	F1	78.85	98.33	96.77	91.74	90.52	87.27	95.93	94.17	87.13	88.00	90.87
	AUROC	92.26	98.66	98.15	94.45	96.38	94.83	98.16	97.41	96.40	94.73	96.14
UNITE† [CVPR 2025]	R	96.29	100.00	98.69	96.93	99.86	94.62	100.00	98.12	92.11	89.89	89.60
	ACC	-	-	-	-	-	-	-	-	-	-	-
	F1	-	-	-	-	-	-	-	-	-	-	-
	AUROC	-	-	-	-	-	-	-	-	-	-	-
D³‡ [ICCV 2025]	R	92.00	92.00	91.80	92.00	92.00	92.00	92.00	92.00	89.29	92.00	91.71
	ACC	89.00	93.10	94.90	91.30	93.70	95.60	93.60	90.70	92.86	83.40	91.82
	F1	89.32	93.02	94.74	91.36	93.59	95.44	93.50	90.82	92.59	84.71	91.91
	AUROC	95.18	96.91	98.54	96.74	97.75	98.74	97.82	96.42	96.78	91.35	96.62
Skyra† [arXiv 2025]	R	66.29	95.86	97.92	82.71	72.00	96.23	98.07	94.29	95.64	78.63	87.66
	ACC	79.93	94.43	96.09	88.50	83.50	95.25	95.92	94.32	95.54	86.56	91.00
	F1	76.76	94.51	96.16	87.79	81.36	95.30	96.01	94.32	95.50	85.41	90.00
	AUROC	-	-	-	-	-	-	-	-	-	-	-
Ours‡	R	94.12	96.38	96.21	93.47	94.63	95.05	96.44	94.18	92.36	95.27	94.81
	ACC	94.31	95.22	96.08	93.14	95.47	96.12	95.33	93.27	93.05	94.18	94.62
	F1	94.71	95.56	96.44	92.88	94.19	96.25	95.17	93.11	93.24	94.36	94.59
	AUROC	98.32	98.74	98.47	96.91	97.63	98.02	98.71	97.44	97.16	96.95	97.84

Table 2. Comparison of false-detection performance. FPR, FNR, and EER are reported in percentages, with lower values indicating fewer false alarms, fewer missed detections, and a lower balanced detection error, respectively.

Method	FPR↓	FNR↓	EER↓
NSG-VD	11.51	5.09	7.74
D³	8.00	8.93	8.39
Ours	5.66	4.87	5.27

$\mathcal{G}_r^+$, incorporates the supported revisions from $\mathcal{G}_r^-$, and synthesizes the updated ERG $\mathcal{G}_{r+1}$ while determining whether another deliberation round is required. If another round is required, the updated graph becomes the current ERG for the next round. Deliberation terminates when no revision is accepted and no additional evidence is requested, or when the maximum number of deliberation rounds is reached, at which point $\mathcal{R}_r^*$ is returned as the final result.

Together, the three deliberators form a closed refinement loop that repeatedly audits and updates the evidential basis

of the current authenticity judgment. As shown in Fig. 3, iterative graph updates progressively expose and correct unsupported or conflicting evidence and relations across rounds, reducing their influence on the final decision and thereby improving detection reliability.

4. Experiments

In this section, we empirically evaluate Hermes. We first introduce the experimental setup and then report comparisons with state-of-the-art methods, false-detection analysis, and ablation studies.

4.1. Experiment Setup

Dataset and Evaluation Protocol. To comprehensively evaluate detection performance and cross-generator generalization, we conduct experiments on the GenVideo benchmark (Chen et al., 2024). GenVideo is a large-scale benchmark for AI-generated video detection that includes ten

Table 3. **Ablation studies of Hermes.** We examine the contributions of the core components and the effects of the number of deliberation rounds and the RAG top- k , reporting AUROC and FNR.

(a) Core component ablation.			(b) Effect of deliberation rounds.			(c) Effect of RAG top- k .		
Module	AUROC $\uparrow$	FNR $\downarrow$	Rounds	AUROC $\uparrow$	FNR $\downarrow$	Top- k	AUROC $\uparrow$	FNR $\downarrow$
w/o RAG	95.12	6.83	1	96.83	5.41	1	96.03	6.42
w/o ERG	95.94	6.21	2	97.12	5.34	3	97.36	5.12
w/o Deliberation	96.21	5.98	3	97.84	4.87	5	97.84	4.87
Ours	97.84	4.87	4	97.79	4.92	7	97.21	5.31

generated-video subsets spanning diverse video generation methods. Hermes requires no task-specific training and is directly evaluated on the official test set. Unless otherwise specified, the training set for training-based baselines consists of real videos from Kinetics-400 (Kay et al., 2017) and generated videos from Pika (pik), following prior work (Zhang et al., 2025c). All methods are evaluated on the same official test set for fair comparison.

Evaluation Metrics. Following prior work (Zhang et al., 2025c), we evaluate performance using recall, accuracy, F1 score, and AUROC as the primary metrics. We additionally report the false positive rate (FPR), false negative rate (FNR), and equal error rate (EER) to provide a more detailed analysis of detection errors.

Baseline Methods. To provide a comprehensive evaluation, we compare Hermes with a diverse set of state-of-the-art methods, including TALL (Xu et al., 2023), NPR (Tan et al., 2024), STIL (Gu et al., 2021), DeMamba (Chen et al., 2024), UNITE (Kundu et al., 2025), D³ (Zheng et al., 2025), NSG-VD (Zhang et al., 2025c), and Skyra (Li et al., 2025c).

Implementation Details. We build the agentic workflow with LangGraph (LANGCHAIN, 2025), and perform local model inference and forensic tool execution using PyTorch (Paszke et al., 2019) on the NVIDIA A800 GPU. We use Qwen3-VL-8B as the VLM for perceptual context analysis (Φ_{VLM}), initial reasoning (Φ_{ini}), and all three deliberator roles (Φ_{pri} , Φ_{cnt} , and Φ_{cons}), and ChatGPT-5 as the MLLM (Φ_{MLLM}) for both offline knowledge-base construction and inference-time strategy planning. To improve inference efficiency, each input video is uniformly sampled at 1 fps, and each group of five sampled frames forms a video clip. The Forensic Knowledge Base contains $M = 19$ evidence-tool pairs corresponding to 10 temporal and 9 spatial evidence types. For each input video, we retrieve the top- k knowledge entries from the Forensic Knowledge Base, where $k = 5$. Each clip node in the initial ERG is connected to at most three evidence nodes, and the maximum number of deliberation rounds is set to $N = 3$.

4.2. Comparison with State-of-the-Art Methods

Overall Detection Performance. Table 1 reports the detection results on GenVideo. Hermes achieves the highest aver-

age performance across all four metrics, reaching 94.81% Recall, 94.62% ACC, 94.59% F1, and 97.84% AUROC. Compared with the strongest overall baseline, D³, Hermes delivers an average gain of 2.45 percentage points across the four metrics. Notably, despite requiring no task-specific training, Hermes surpasses the fully supervised baselines in overall performance. Relative to NSG-VD, Hermes improves F1 and AUROC by 3.72 and 1.70 percentage points, respectively. Hermes also maintains consistently strong performance across diverse generation sources, demonstrating robust cross-generator generalization. This generalization benefits from the coordination of three components: video-conditioned planning adapts forensic checks to each input, ERG-based reasoning organizes evidence into an explicit and traceable structure, and graph-grounded deliberation iteratively supplements, challenges, and reconciles evidence. Together, these components reduce reliance on fixed, generator-specific artifacts and enable more stable authenticity judgments under cross-generator distribution shifts.

False Detection Analysis. Table 2 compares the false positive rate (FPR), false negative rate (FNR), and equal error rate (EER), where lower values indicate fewer false alarms, fewer missed detections, and a lower balanced detection error, respectively. Hermes achieves the lowest values across all three metrics. Compared with D³, Hermes reduces FPR, FNR, and EER by 2.34, 4.06, and 3.12 percentage points, respectively, demonstrating improved control over both false alarms and missed detections. Graph-grounded deliberation contributes to this reliability by supplementing missing evidence, challenging questionable findings and relations, and adjudicating proposed revisions before incorporating them into the updated ERG. As illustrated in Figure 3, evidence nodes and relations are iteratively added, revised, or removed according to their evidential support, preventing weakly supported cues from exerting disproportionate influence on the final judgment.

4.3. Ablation Studies

To assess the contributions of the core components and the effects of key settings, we conduct comprehensive ablation studies on GenVideo, with AUROC and FNR reported in Table 3.

Effect of Core Components. Table 3a evaluates the contributions of the three core components in Hermes: RAG-based strategy planning, ERG-based evidence organization, and graph-grounded deliberation. Removing RAG causes the largest performance degradation, decreasing AUROC by 2.72 percentage points and increasing FNR by 1.96 percentage points. This result shows that retrieving video-relevant forensic knowledge is critical for composing detection strategies tailored to diverse content and generation artifacts. Removing the ERG also degrades both metrics, confirming that explicitly organizing evidence items and their relations is important for integrating clip-level observations into a globally grounded representation. Similarly, disabling graph-grounded deliberation leads to inferior performance, demonstrating the benefit of supplementing missing evidence and examining unreliable nodes and inconsistent relations before the final decision. Overall, the full configuration consistently performs best, indicating that the three components make complementary contributions to detection performance and reliability.

Effect of Deliberation Rounds. Table 3b examines the effect of varying the number of deliberation rounds. Performance improves progressively from one to three rounds, as additional iterations allow the ERG to be further supplemented and refined. Three rounds achieve the best AUROC and FNR, whereas increasing the number to four brings no further improvement and incurs additional inference cost. We therefore adopt three rounds as the default setting, offering the best balance between performance and efficiency among the evaluated configurations.

Effect of Retrieved Knowledge Size. Table 3c examines the effect of the number of knowledge entries retrieved in the RAG stage. Increasing k from 1 to 5 progressively improves AUROC and reduces FNR, indicating that broader forensic knowledge coverage helps the planner formulate more targeted analysis strategies. When k is increased to 7, performance declines, suggesting that redundant or weakly relevant entries may dilute useful guidance and interfere with strategy planning. We therefore adopt $k=5$ as the default retrieval setting.

5. Conclusion

We present **Hermes**, an evidence-driven agentic framework that reformulates AI-generated video detection as a structured, video-conditioned forensic reasoning process. By integrating Adaptive Instance-Conditioned Detection Strategy Planning, Evidence-Centric Reasoning and Verification, and Graph-Grounded Evidence Deliberation, Hermes adapts forensic analysis to each video, organizes heterogeneous evidence within an Evidence Reasoning Graph, and couples authenticity assessment with structured and verifiable forensic explanations. Extensive experiments

on GenVideo demonstrate state-of-the-art detection performance and strong cross-generator generalization without task-specific training. We hope this work offers new insights into leveraging multimodal foundation models for evidence-grounded and explainable video forensics.

Impact Statement

This work advances trustworthy synthetic-media analysis by proposing an evidence-driven framework for AI-generated video detection. By improving detection reliability, interpretability, and auditability, Hermes can support media verification and help mitigate malicious uses of generative video models. However, the disclosed forensic cues may potentially be exploited to improve generation systems or evade existing detectors. To reduce this risk, we will adopt responsible release practices and appropriate access control for sensitive implementation details. Overall, we encourage the research community to maximize the benefits of reliable AIGV detection while minimizing its potential misuse.

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